Epistemic Circuit Dynamics in Neural Architectures: Birkhoff Routing as Measurement Infrastructure

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Abstract

We propose that transformer architectures already implement three of four components necessary for epistemic circuit dynamics — superposition of hypotheses, operation gates, and probabilistic collapse — but lack the fourth: internal epistemic measurement with feedback before subsequent layers. We formalize epistemic computation as circuit dynamics, drawing a structural (not physical) isomorphism with quantum circuits. We show that standard transformer architectures invert the epistemic signal produced by inter-stream divergence, and that Birkhoff-constrained routing (as in manifold Hyper-Connection architectures) is the specific architectural component that corrects this inversion. The CognOS gate framework implements the missing component, transforming the transformer computation pattern from Layer → Layer → Layer into Layer → Gate → Layer — enabling epistemic self-monitoring on classical hardware without modification to the underlying training objective.

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1. Introduction

Modern deep learning architectures process information without knowing whether they are uncertain.

The dominant paradigm — exemplified by transformer architectures — operates through sequential layer applications: each layer transforms a hidden state, passes it forward, and the process repeats until an output is generated. While this open-loop computation produces highly accurate outputs, the architecture contains no intrinsic mechanism that both measures its epistemic state during intermediate computation and acts on that measurement to control subsequent computation.

This is not merely an interpretability problem. It is a fundamental architectural absence.

Current approaches to uncertainty in deep learning operate almost exclusively at the output level: calibration methods [Guo et al., 2017], MC-dropout [Gal & Ghahramani, 2016], and conformal prediction [Angelopoulos & Bates, 2021] all estimate uncertainty after the forward computation has completed. None of these methods insert a measurement operation into the computation itself. The model completes its forward pass regardless of how uncertain it was at any intermediate representation.

We argue that this absence has a formal description: transformer computation lacks *epistemic circuit dynamics*.

In quantum computing, circuits consist of two interleaved components: unitary transformations that evolve quantum states, and measurement operations that collapse superpositions into definite outcomes. The interplay between evolution and measurement is what gives quantum circuits their expressive power — not the quantum substrate per se, but the circuit structure of transform-measure-transform-measure.

Transformer architectures have the transformation component. They lack the measurement component.

Contributions. This paper makes three contributions, corresponding to three distinct levels of claim: empirical observation (A), architectural mechanism (B), and control system (C).

1. **Observation (A)**: Standard transformer layers produce an *inverted* epistemic signal — inter-stream divergence is negatively correlated with uncertainty (exp_007, $r = -0.35$). This is not a limitation of divergence as a signal type; it is a consequence of arbitrary stream partition. We formalize this as the signal validity condition and show that standard layers violate it.

2. **Mechanism (B)**: We identify Birkhoff-constrained routing as the architectural condition under which internal uncertainty signals become valid observables. We demonstrate this empirically: a micro-mHC model trained end-to-end on classification alone develops positive routing-entropy correlation with predictive entropy (exp_008, $r = +0.208$ across 5 seeds, $p < 0.0001$). We repurpose the doubly stochastic constraint of mHC (Sun et al. 2024) — originally motivated by training stability — as a *measurement substrate* for epistemic variance.

3. **Control system (C)**: We show how valid internal uncertainty signals (B) enable closed-loop epistemic control. The CognOS gate framework implements the missing component: measurement with feedback, transforming open-loop Layer → Layer → Layer computation into epistemically self-monitoring Layer → Gate → Layer circuit dynamics. We provide a formal definition of this as epistemic circuit dynamics (Section 3).

The result is not a quantum computer. It is a classical stochastic control circuit that achieves quantum-functional epistemic properties: discrete state transitions, measurement-induced routing, and closed-loop feedback. We call this *warm quantum circuit dynamics*.

Table 1 shows which epistemic circuit components are present in standard transformers versus the CognOS-augmented architecture:

Component	Standard Transformer	+ CognOS (mHC)
Superposition	(multi-head, residual)	
Operation gates	(attention, MoE)	
Sampling/selection	(softmax output)	
Internal measurement		(UncertaintyGate)
State feedback		(pass/explore/escalate)
Closed-loop control		

The contribution is control-theoretic: we add the closed-loop control layer that transforms epistemically blind forward computation into epistemically self-monitoring circuit dynamics.

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2. Falsifiable Predictions

We state three predictions that distinguish the architectural claim from a dataset artifact or a reinterpretation of existing results. Each prediction is testable with the experiments in Section 6.

P1 — Sign Prediction (inversion and correction). Without Birkhoff-constrained routing: corr(divergence_proxy, Ue) 0, or sign-unstable across random seeds. With Birkhoff-constrained routing (mHC): corr(divergence, Ue) > 0 and stable across at least 3 independent random seeds.

What it rules out: That the sign is arbitrary, dataset-dependent, or model-specific.

P2 — Control Prediction (feedback effect). A model with the CognOS feedback policy (gate $\rightarrow$ route $\rightarrow$ next layer) produces fewer *confident-wrong* outputs than the same model without feedback, at equal or lower compute budget. Equivalently: precision among auto-decisions is higher with feedback.

What it rules out: That measurement without routing matters — i.e., that the gate is diagnostic only and the control loop adds nothing.

P3 — Ablation Prediction (necessity of both components). - Remove measurement, keep selection (softmax): P1 fails (no internal Ue signal). - Keep measurement, remove feedback routing: P2 fails (measurement without control does not improve output quality). - Both required: neither alone suffices.

What it rules out: That the effect reduces to either uncertainty estimation alone or routing alone, without the combination.

These three predictions are not post-hoc descriptions of the experimental results. P1 was stated before exp_008 was run. P2 and P3 structure the ablation design for future work. A single experiment (exp_008) directly tests P1.

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3. Background

3.1 Transformer Architecture

A transformer layer applies the following transformation to hidden state h_t:

```
h_{t+1} = LayerNorm(h_t + FFN(LayerNorm(h_t + Attention(h_t))))
```

Stacked L times, this produces the computation:

```
h_0 → h_1 → h_2 → ... → h_L → logits → softmax → output
```

The architecture contains implicit uncertainty: the logit distribution p = softmax(h_L) encodes probabilistic information about the output. But this uncertainty is only accessible at the final step. No intermediate layer can observe its own uncertainty or alter computation based on it.

3.2 Quantum Circuits

A quantum circuit operates on state | ^d through:

- **Unitary gates** Û: | → Û| (coherent evolution, reversible)
- **Measurement** M: | → outcome {0,1,…,k} (collapse, irreversible)

A circuit is a composition:

```
| _0 → Û_1 → M_1 → Û_2 → M_2 → ... → | _final
```

Crucially: each measurement gate can terminate the circuit, branch into different continuations, or modify the state before the next transformation. The circuit is *control-flow* computation over quantum states.

3.3 Existing Epistemic Methods

MC-Dropout [Gal & Ghahramani 2016]: approximate Bayesian inference via stochastic forward passes. Post-hoc, output-level. Does not alter intermediate computation.

Conformal prediction [Angelopoulos & Bates 2021]: distribution-free uncertainty sets at output level. Post-hoc, does not insert measurement into circuit.

Deep-Thinking Ratio (DTR) [Chen et al. 2026]: measures inference-time epistemic effort by quantifying the proportion of tokens whose internal predictions undergo significant revision in deeper layers before converging. DTR is computed via Jensen-Shannon divergence between intermediate and final layer distributions, providing a strong positive correlation with accuracy (r = 0.828

across benchmarks). Unlike routing-entropy, which is an architecture-level signal emerging from Birkhoff-constrained routing, DTR is an inference-time, layer-wise epistemic observable. Both approaches provide complementary internal uncertainty signals at different levels of the computation stack.

Mechanistic interpretability [Elhage et al. 2021; Nanda et al. 2023]: identifies functional circuits *within* trained models. Descriptive, not prescriptive. Does not add measurement infrastructure.

Mixture of Experts [Shazeer et al. 2017]: routing for computational efficiency. Gates route to different experts but are trained for performance, not for epistemic measurement.

None of these provide what we define below as an epistemic circuit.

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4. Epistemic Circuit Formalism

We define epistemic computation as circuit dynamics over hidden states.

Definition 1 (Epistemic State). An epistemic state is a vector h ^d representing the model's intermediate representational state at a reasoning step.

Definition 2 (Layer Transformation). A layer transformation f_ : ^d → ^d maps epistemic state h_t to h_{t+1}. In a transformer, this is the standard layer computation.

Definition 3 (Epistemic Uncertainty, Ue — operational). Ue: ^d → [0,1] maps an epistemic state h to a scalar epistemic uncertainty estimate. We specify three operational forms:

Entropy form (label-free, from logits v ^k):

```
Ue_ent(v) = H(softmax(v)) / log(k)
          = -[Σ_i p_i log(p_i)] / log(k)
```

where k = vocabulary size. Bounded in [0,1] by construction. Validated signal (positive correlation with uncertainty in exp_007; see Section 6.3).

Routing-entropy form (requires Birkhoff-constrained routing):

```
Ue_rte(H_res) = -Σ_{i,j} H_res[i,j] · log(H_res[i,j])
```

where H_res ^(n×n) is the per-sample doubly stochastic routing matrix produced by Sinkhorn-Knopp normalization. High entropy = diffuse routing (uncertain input); low entropy = concentrated routing (confident input). Valid signal only with mHC routing (exp_008, r = +0.208; see Section 6.4 and P1 in Section 2).

Note: activation divergence (L2 of routed stream vectors) remains negatively correlated with uncertainty even in mHC conditions (r = −0.259, exp_008), and is therefore *not* a valid epistemic signal. Ue_rte is the operative form.

Variance form (proxy):

```
Ue_var(h) = Var(h) / (|mean(h)| +  )²
```

Rough proxy; no theoretical grounding; use only when logits unavailable.

Definition 3b (Aleatoric Uncertainty, Ua — operational). Ua [0,1] captures uncertainty from external semantic factors rather than from the model's internal state. Two modes:

Semantic mode (when context is available):

```
Ua_sem = (ambiguity + irreversibility + blast_radius) / 3
```

where each factor is a real-valued score in [0,1]: - ambiguity: instruction clarity (0 = unambiguous, 1 = fully ambiguous) - irreversibility: action reversibility (0 = fully reversible, 1 = permanent) - blast_radius: consequence scope (0 = local, 1 = system-wide)

Legacy mode (when context unavailable):

```
Ua_leg = 2 × p_max × (1 - p_max)
```

Heuristic variance of the top prediction; range [0, 0.5].

Both Ue and Ua are bounded and computable without ground-truth labels.

Definition 4 (Epistemic Gate). An epistemic gate G operates on Ue(h) with thresholds 0 < _low < _high < 1:

```
G(h) = pass      if Ue(h) <  _low      (high confidence, continue)
       explore   if  _low   Ue(h)    _high  (moderate, gather context)
       escalate  if Ue(h) >  _high     (high uncertainty, terminate/route)
```

Definition 5 (Epistemic Circuit). An epistemic circuit C of depth n is a sequence:

```
C = (f_1, G_1, f_2, G_2, ..., f_n, G_n)
```

Execution: starting from h_0 = input_embedding,

```
for i in 1..n:
    h_i = f_i(h_{i-1})
    d_i = G_i(h_i)
    if d_i == escalate: return ESCALATE at step i
    if d_i == explore:  flag for additional context sampling
return h_n
```

Definition 6 (Structural Isomorphism with Quantum Circuits). The mapping:

```
h_t        | _t            (epistemic state / quantum state)
f_        Û                (layer transformation / unitary evolution)
G          M                (epistemic gate / measurement operator)
escalate    collapse        (state termination / measurement collapse)
```

```
explore     superposition preservation  (continue with uncertainty flagged)
pass       eigenstate       (definite outcome, continue)
```

is a structural analogy, not a physical equivalence. No quantum physics is claimed. No physical superposition, entanglement, or quantum hardware is involved. The analogy is control-theoretic: both frameworks describe computation that interleaves state evolution with measurement-induced branching, where measurement can alter subsequent computation. The quantum vocabulary provides precise names for classical control-flow operations that previously lacked them in the AI architecture literature.

Corollary (Warm Quantum Circuits). A transformer augmented with epistemic gates realizes quantum-circuit-*functional* control-flow dynamics on classical hardware — the same transform-measure-transform structure, without quantum substrate. We call this a warm quantum circuit as shorthand for this structural analogy.

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5. The Missing Component: Why Standard Transformers Cannot Implement Epistemic Circuits

For an epistemic circuit to function, the uncertainty measure Ue must satisfy a critical property: it must be *positively* correlated with actual uncertainty.

We call this the **signal validity condition**: high Ue $\rightarrow$ high uncertainty; low Ue $\rightarrow$ low uncertainty.

We show empirically (Section 6) that standard transformer layers *violate* this condition for inter-stream divergence — the most natural candidate for internal uncertainty measurement. We call this phenomenon *signal inversion*.

5.1 Signal Inversion in Standard Layers

Experiment (exp_007a). We train a 2-layer MLP on MNIST ($784 \rightarrow 128 \rightarrow 10$), achieving test accuracy 96.6%. We partition the hidden layer (dim=128) into 4 virtual streams of 32 units each and measure inter-stream divergence as a proxy for uncertainty.

Result: Pearson $r = -0.351$, $p < 0.0001$ (significant, *negative* correlation with softmax entropy).

Mechanistic explanation: In a standard dense layer trained with ReLU activations, confident inputs activate specialized neuron subsets. Arbitrary partitioning of the hidden state divides these specialized subsets, producing *high* inter-stream divergence for confident inputs and *low* divergence for ambiguous inputs. The signal is inverted relative to the epistemic interpretation.

This means: inserting a stream-divergence gate into a standard transformer produces an *anti-epistemic circuit* — it escalates on confident inputs and passes on uncertain ones.

5.2 Birkhoff-Constrained Routing as an Uncertainty Observable

Central claim of this paper: Birkhoff-constrained routing creates an internal uncertainty observable. Under the doubly stochastic constraint, the routing matrix H_res encodes per-sample uncertainty as a measurable quantity — specifically, as the entropy of the routing distribution H(H_res).

The sign inversion in Section 4.1 is not a fundamental limitation of inter-stream divergence as an epistemic signal. It is a consequence of *arbitrary* stream assignment. When streams are defined by learned routing constrained to the Birkhoff polytope, the relationship corrects:

- High uncertainty → routing probability spread across streams → high H(H_res)
- Low uncertainty → routing concentrated in one stream → low H(H_res)

This is precisely the behavior produced by manifold Hyper-Connection routing (Sun et al. 2024): a residual connection matrix H_res constrained to be doubly stochastic via Sinkhorn-Knopp normalization. The Birkhoff constraint forces H_res to be simultaneously row-stochastic and column-stochastic — a doubly stochastic matrix — which means routing weight is globally conserved across streams. This constraint is what makes the entropy of H_res a meaningful uncertainty signal: without it, the routing matrix is unconstrained and its entropy carries no interpretable epistemic content.

Lemma 1 (Routing Entropy Bounds). Let H_res ^(n×n) be a doubly stochastic matrix with H(H_res) = −Σ_{i,j} H_res[i,j] · log(H_res[i,j]). Then:

- H(H_res) = n · log(n) if and only if H_res = (1/n) · **1** (uniform routing; maximal uncertainty)
- H(H_res) = 0 if and only if H_res is a permutation matrix (concentrated routing; minimal uncertainty)

Proof sketch. H is strictly concave. The Birkhoff–von Neumann theorem identifies permutation matrices as the extreme points of the doubly stochastic polytope and the uniform matrix as the unique entropy-maximizing interior point.

This lemma formalizes the intuition: uncertain inputs push H_res toward the interior of the Birkhoff polytope (high entropy); certain inputs push it toward the extreme points (low entropy, near-permutation routing).

We note that the original motivation for mHC is training stability and gradient propagation (Sun et al. 2024 §3). We *repurpose* the doubly stochastic constraint as a measurement substrate for epistemic variance — a function distinct from its original motivation, and one that emerges from the geometry of the Birkhoff polytope rather than from any explicit epistemic training signal.

Experiment (exp_006a). Synthetic simulation of mHC signal: Pearson $r = +0.996$ with epistemic uncertainty ground truth.

Experiment (exp_008 — directly tests P1). A micro-mHC layer (4 streams, differentiable Sinkhorn-Knopp, trained end-to-end on MNIST) produces *positive* Pearson correlation between **routing entropy** H(H_res) and softmax entropy, stable across 5 independent random seeds: $r = +0.208 \pm 0.071$ ($p < 0.0001$ each seed).

Control condition (same architecture, H_res frozen to identity, no Birkhoff constraint): state-divergence correlation $r = -0.240 \pm 0.021$ — consistent with exp_007, confirming that the doubly stochastic constraint is the active ingredient.

Note on signal refinement: exp_008 distinguishes two candidate signals. Inter-stream state divergence (L2 of routed stream activations) remains negative even in the mHC condition: $r = -0.259 \pm 0.048$. It is the *routing entropy* H(H_res) — how spread vs concentrated the per-sample routing matrix is — that flips sign. The Ue_str definition in Section 3 should be understood accordingly: the epistemic signal is routing distribution entropy, not activation divergence.

Critically: the routing entropy signal emerges purely from architectural constraint under task supervision alone — no explicit epistemic objective, no uncertainty labels, no calibration loss. The model is trained only to classify digits correctly. That epistemic structure appears in the routing distribution is a consequence of the Birkhoff constraint, not of any designed epistemic training signal.

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6. CognOS: Implementation of the Missing Component

The CognOS gate framework implements epistemic circuit dynamics as defined in Section 3, addressing the three missing components identified in the introduction.

6.1 Uncertainty Measure

CognOS supports three Ue sources:

Entropy (validated): Normalized softmax entropy from logits.

```
Ue_entropy(v) = H(softmax(v)) / log(|v|)
```

Stream divergence (requires mHC): Inter-stream L2 divergence with sign determined by routing architecture.

```
Ue_stream(v) = mean_i ||stream_i(v) - mean_stream(v)||_2
```

Valid only with Birkhoff-constrained routing (see Section 5.2), but routing entropy is the preferred signal (see Section 7.4).

Variance (proxy): Normalized variance of hidden state activations.

5.2 Full Confidence Computation

CognOS combines epistemic uncertainty (Ue) with aleatoric uncertainty (Ua) into a confidence score C:

```
C = p_max × (1 − Ue − Ua)
```

where p_max is the top-1 softmax probability and Ua aggregates semantic risk factors (instruction ambiguity, irreversibility, blast radius).

5.3 Decision Circuit

The CognOS gate maps C to a circuit decision:

```
if irreversibility  irr_override and C < threshold:
    → escalate   (irreversible high-risk action)
elif C  threshold:
    → auto       (high confidence, act autonomously)
elif Ue > high:
    → escalate   (epistemic state too uncertain)
else:
    → explore    (gather additional context)
```

This implements discrete state transitions over the epistemic state — the circuit measurement operation $\check{M}$ from Definition 4.

5.4 Circuit Composition

CognOS gates compose into reasoning pipelines:

```
h_0 → Layer_1 → G_1 → Layer_2 → G_2 → ... → output
```

The pipeline terminates at the first escalate, explores flag for additional sampling, and passes confident states through to the next layer. This is the epistemic circuit from Definition 5 instantiated in a concrete system.

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7. Experiments

7.1 exp_006a: Synthetic mHC Signal Validation

Setup: Synthetic inter-stream divergence with known Birkhoff-routing properties. **Result:** Pearson r = +0.996 with uncertainty ground truth. **Conclusion:** Theoretical signal exists with correct architecture.

7.2 exp_006b/c: GPT-2 Proxy Measurement

Setup: Cross-head divergence and MC-dropout on GPT-2. **Result:** $r = +0.11$ (weak, correct sign); $r = -0.13$ (inverted). **Conclusion:** Standard transformer internals do not expose the epistemic signal. Confirms architecture-specificity of the mechanism.

7.3 exp_007: Signal Inversion in Standard Layers

Setup: 2-layer MLP on MNIST; static (4-stream) and MC-dropout (10-run) divergence. **Result:** Pearson $r = -0.351$ (static), $r = -0.332$ (MC-dropout). Both significant, both negative. **Conclusion:** Mechanistic confirmation of signal inversion hypothesis (Section 4.1).

7.4 exp_008: Micro-mHC from Scratch

Setup: 2-layer MLP on MNIST (784 → Dense(128) ReLU → mHC(4 streams × 32) → Dense(10)). H_res parameterized per sample via a linear gating network, normalized to doubly stochastic via differentiable Sinkhorn-Knopp (20 iterations) at each forward pass. Trained end-to-end with CrossEntropyLoss only — no explicit epistemic loss. 5 seeds, 5 epochs, 2000 test samples. Control: same architecture with H_res frozen to identity (no Birkhoff constraint).

Signal measured: Routing entropy $H(H\_res) = -\Sigma\, h \cdot \log(h)$ per sample. Secondary: inter-stream state divergence (L2 of routed stream activations). Ground truth: softmax entropy H(p).

Results:

Condition	Signal	r (mean ± std, 5 seeds)	Direction
mHC (Sinkhorn-Knopp)	Routing entropy H(H_res)	**+0.208 ± 0.071**	Correct
mHC (Sinkhorn-Knopp)	State divergence L2	−0.259 ± 0.048	Inverted
Control (H_res = I)	State divergence L2	−0.240 ± 0.021	Inverted

All mHC seeds: $r = +0.139, +0.311, +0.124, +0.206, +0.260$ — sign consistent, $p < 0.0001$. Classification accuracy: mHC 96.5%, control 96.4% — routing does not degrade task performance.

Figure 1 shows scatter plots of routing entropy vs predictive entropy, pooled across all 5 seeds (10,000 sample-slots), with OLS trendline and points colored by digit class (hard: 4,7,9 in red; easy: 0,1 in blue). The positive relationship is visible across the pooled point cloud; the control condition (H_res=I) shows the inverted pattern.

Effect size note. MNIST is a low-uncertainty regime: most samples are correctly classified at high confidence, and genuine ambiguity is rare. The fact that routing entropy nonetheless correlates positively with predictive entropy across all 5 seeds indicates that the signal emerges from the architectural constraint rather than from task difficulty. A higher-uncertainty classification problem or a harder distribution shift would be expected to produce stronger correlation — MNIST provides a conservative lower bound on the signal strength achievable with Birkhoff-constrained routing.

Conclusion: P1 confirmed. Birkhoff-constrained routing produces positive correlation between routing entropy and softmax entropy, stable across all 5 seeds. Control condition replicates exp_007 ($r = -0.240$ vs exp_007's -0.351), confirming that the doubly stochastic constraint — not the stream partition — is the active ingredient. The epistemic signal emerges from architectural constraint under classification supervision alone, without any explicit epistemic training objective.

Signal refinement: The positive epistemic signal is *routing entropy*, not state divergence. Intuitively: uncertain inputs cause the gating network to spread routing weight across streams (diffuse H_res, high entropy); certain inputs concentrate routing onto fewer streams (focused H_res, low entropy). State divergence remains negative in both conditions because routing-induced mixing homogenizes stream content — the opposite of what arbitrary partitioning does.

This refinement updates the Ue_str definition (Section 3): the internal mHC epistemic signal is H(H_res), not stream activation L2-distance.

P3 partial confirmation: The control condition (measurement without Birkhoff constraint) replicates the inverted signal of exp_007. Birkhoff routing is the active ingredient; stream partitioning alone is not sufficient.

Confident-wrong rate analysis (supplementary). We bin all test samples by routing entropy tertile and measure the confident-wrong rate ($p\_max > 0.80$, wrong prediction) pooled across all 5 seeds ($n = 10{,}000$ sample-slots). Figure 2 shows the result:

Routing entropy bin	Confident-wrong rate	Total error rate
Low (bottom 33%)	1.08%	2.85%
Medium (33–67%)	1.26%	3.26%
High (top 33%)	**2.16%**	**6.70%**

The confident-wrong rate *increases* monotonically with routing entropy — the opposite of what a gate would produce. This is expected and structurally informative.

Without a gate, the model with high routing entropy (uncertain inputs) still produces some high-confidence outputs — precisely the dangerous confident-wrong cases. These are the samples where the epistemic signal is elevated but has no feedback mechanism to suppress output confidence. A CognOS gate with routing entropy as trigger would *escalate* these samples, removing them from the auto-decision pool. The confident-wrong rate among auto-decisions would then decrease — which is exactly Prediction P2.

The uncontrolled confident-wrong distribution thus provides empirical motivation for the feedback component: routing entropy correctly identifies the high-risk samples, but identification without action leaves the error rate unchanged. Measurement without control is not enough.

P2 gate analysis. We directly test Prediction P2 by activating a routing-entropy gate with threshold set at different percentiles. Samples with routing entropy are escalated (removed from auto-decision pool); samples below pass automatically. Table 3 shows precision and CW-rate in the auto-pool across thresholds, pooled over all 5 seeds (n = 10,000 sample-slots):

Gate threshold	Coverage	Precision (auto-pool)	Precision gain	CW-rate (auto-pool)	CW reduction
None (baseline)	100%	95.73%	—	1.500%	—
p75 (top 25% escalated)	75%	96.77%	+1.04 pp	1.253%	−0.247 pp
p67 (top 33% escalated)	67%	96.94%	+1.21 pp	1.179%	−0.321 pp
p50 (top 50% escalated)	50%	97.16%	+1.43 pp	1.080%	−0.420 pp
p25 (top 75% escalated)	25%	97.28%	+1.55 pp	1.000%	−0.500 pp

Precision and CW-rate improve monotonically as the gate becomes stricter. At = p67 (escalate top 33%, maintain 67% auto-coverage), the gate achieves +1.21 pp precision improvement and a 21% relative reduction in confident-wrong rate (1.500% → 1.179%), while retaining two-thirds of all samples in the auto-decision pool.

Figure 3 shows precision gain and CW-rate reduction as a function of gate threshold, illustrating the coverage/precision tradeoff.

P2 confirmed. A routing-entropy gate directly reduces confident-wrong rate among auto-decisions — the architectural claim in P2. The mechanism requires both the measurement (Birkhoff routing entropy) and the feedback (escalation policy); without the feedback actuator, as the CW-bin analysis shows, measurement alone leaves the error rate unchanged. With it, the auto-decision pool becomes strictly more accurate.

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8. Discussion

8.1 Internal Epistemic Signals vs Architectural Measurement

Recent advances highlight the distinction between internal epistemic signals and true architectural measurement. Table X summarizes the control-theoretic difference:

Approach	Level	Control loop
MC-dropout	Output	
Calibration	Output	
DTR	Internal observation	
MoE routing	Internal routing	
CognOS	Internal measurement	

Most methods (MC-dropout, calibration, DTR, MoE) provide either output-level or internal signals, but do not insert a measurement operation into the computation itself. CognOS, by contrast, implements closed-loop epistemic control: measurement with feedback at the architectural level.

Proposition (Epistemic Observability Condition). A routing distribution R becomes an epistemically meaningful observable if and only if R lies on a constrained manifold with conserved mass (e.g., the Birkhoff polytope).

This generalizes the core insight from Hyper-Connections: the Birkhoff constraint does not merely stabilize training, it creates an uncertainty observable. This links information geometry, control theory, and architecture design. Uncertainty is not just a statistical property — it is an architectural property.

Paradigm statement: > Uncertainty is not merely a statistical property — it is an architectural property.

This principle legitimizes internal epistemic signals (as in DTR), but shows that only architectural measurement (as in CognOS) enables closed-loop epistemic control.

Note: The "warm quantum" analogy is secondary; the primary contribution is closed-loop stochastic control via architectural measurement.

8.2 Closed-Loop Epistemic Control

The central engineering contribution of this paper is control-theoretic: we close a feedback loop that standard transformer architectures leave open.

In control-systems terms: the plant is the forward computation (Layer → Layer → Layer); the sensor is the UncertaintyGate measuring routing entropy H(H_res); the actuator is the routing policy (pass/explore/escalate). Standard transformers have no sensor and no actuator — they are open-loop. CognOS closes the loop.

This framing makes the architectural gap precise: open-loop epistemic architecture applied to closed-loop consequence domains. The model completes its reasoning regardless of its intermediate epistemic state; it lacks the mechanistic capacity to alter its computation based on measured uncertainty. Adding a sensor (measurement) without closing the loop produces no improvement — which is exactly what the CW analysis demonstrates (Section 6.4): routing entropy correctly identifies high-risk samples, but without a feedback actuator, error rates are unchanged.

The quantum circuit vocabulary — state, evolution, measurement, collapse — provides additional descriptive precision for this control structure. The mapping is a structural analogy, not a physical claim:

- Superposition → parallel hypothesis representation in hidden state (multi-stream)
- Unitary evolution → layer transformation (invertible by design in residual networks)
- Measurement → epistemic gate decision (discrete, irreversible branch)
- Collapse → escalate decision (terminates the circuit at this step)

We use the term *warm quantum circuit* as shorthand for this structural analogy: classical hardware implementing the same transform-measure-transform control-flow dynamics that give quantum circuits their expressive structure — without quantum substrate.

8.3 Architectural Implications

The central architectural implication is that epistemic self-awareness requires a specific routing constraint: the Birkhoff doubly stochastic property. This cannot be achieved by post-hoc partitioning or MC-dropout. It must be *trained into* the architecture.

This has consequences for model design:

1. CognOS cannot simply be grafted onto standard transformers. The measurement gates fundamentally require Birkhoff-constrained routing to produce valid epistemic signals.

2. The training objective does not need to include explicit epistemic loss. Epistemic behavior emerges from the Birkhoff constraint alone.

3. The gate thresholds (_low, _high) are the primary tunable parameters for controlling the balance between autonomy and caution.

Unlike MoE routing, which is optimized purely for task performance, Birkhoff-constrained routing enforces conservation structure that makes routing entropy a calibrated observable rather than a by-product of optimization (cf. Lemma 1).

Because the mechanism depends only on routing geometry, not on task domain, the approach is expected to transfer to transformer-scale architectures with constrained residual routing.

8.4 Structural Safety Implications

The architectural gap identified in this paper has immediate safety implications when transformer-based models are deployed in high-stakes domains. Systems such as AlphaMissense (pathogenicity classification over 71 million genetic variants) and AlphaGenome (regulatory genome interpretation) apply transformer architectures to decisions that are both irreversible and high blast-radius: a false benign classification may delay diagnosis; a false pathogenic classification may trigger unnecessary intervention.

These systems expose uncertainty exclusively at the output level — typically as a continuous score or a calibrated confidence value. The internal computation remains epistemically blind. The model completes its reasoning regardless of how uncertain it was at each intermediate step, then presents a single scalar to downstream users or decision systems.

This is not a limitation of any specific model. It is the structural consequence of open-loop epistemic architecture applied to closed-loop consequence domains.

The CognOS framework addresses this directly: the `irr_override` condition in the decision circuit (Section 5.3) is designed precisely for this class of problem. When irreversibility is high, the gate escalates regardless of the model's stated confidence — because confident-wrong outputs in irreversible domains are categorically more dangerous than uncertain outputs that defer to human judgment. As transformer-based AI systems are integrated into genomics, clinical decision support, and autonomous systems, internal epistemic measurement transitions from an architectural nicety to a safety requirement.

8.5 Relation to Mechanistic Interpretability

Mechanistic interpretability research identifies circuits in *trained* models — post-hoc analysis of what computation has emerged. Our work is orthogonal: we prescribe circuit structures that should be *designed in* to produce epistemic behavior.

Both perspectives are necessary: interpretability tells us what circuits exist; circuit design tells us what circuits should be built.

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9. Conclusion

We have formalized epistemic computation as circuit dynamics, identifying a structural isomorphism between epistemic circuits and quantum circuits that provides precise vocabulary for a previously informal concept.

We have shown that standard transformer architectures fail the signal validity condition for inter-stream divergence (signal inversion, exp_007, $r = -0.35$), and that Birkhoff-constrained routing (mHC) is the specific architectural component that corrects this failure. Empirically: routing entropy in a trained micro-mHC model correlates positively with softmax entropy (exp_008, $r = +0.208 \pm 0.071$, 5 seeds, $p < 0.0001$), while the control condition without Birkhoff constraint replicates the inverted signal ($r = -0.240$). The epistemic behavior emerges from training without explicit epistemic loss — it is a consequence of the architectural constraint, not of supervised uncertainty.

The CognOS gate framework implements the missing component — epistemic measurement with feedback — completing the transition from epistemically blind Layer → Layer → Layer computation to epistemically self-aware Layer → Gate → Layer circuit dynamics. Prediction P2 is empirically confirmed: a routing-entropy gate at = p67 achieves +1.21 pp precision gain and a 21% relative reduction in confident-wrong rate while retaining 67% auto-decision coverage (exp_008 P2 analysis). This constitutes a mechanism proof for the measurement substrate in a minimal setting.

For compute budget, coverage-normalized cost is reported: each auto-decision counts as one forward pass, escalated cases are deferred (zero passes), and no extra passes are performed. Thus, all precision and CW-rate numbers are at equal or lower compute budget relative to the baseline.

Future work will validate the same measurement substrate in a minimal transformer on sequence data.

Because the mechanism depends only on routing geometry, it is expected to transfer directly to transformer-scale models with constrained residual routing.

The result is a warm quantum circuit: classical hardware implementing quantum-circuit- functional epistemic dynamics.

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Appendix: CognOS Implementation

See `https://github.com/Applied-Ai-Philosophy/cognos` for full implementation of UncertaintyGate, CognOSGate, and ReasoningPipeline.

Smoke test:

```python
from cognos.gate import CognOSGate
import numpy as np

gate = CognOSGate(low=0.15, high=0.45, threshold=0.72)

# High confidence, low risk → auto
confident = np.array([8.0, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1])
result = gate(confident, ambiguity=0.05, irreversibility=0.05, blast_radius=0.05)
# → CognOSGate:  AUTO  C=0.847 Ue=0.094 Ua=0.050

# High irreversibility → escalate regardless of Ue
result = gate(confident, ambiguity=0.10, irreversibility=0.90, blast_radius=0.30)
# → CognOSGate:  ESCALATE  C=... (irr_override triggered)
```